



GE Medical Systems

Technical Publication

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Rev 2**

GE Medical Systems

Mobile CT System Requirements Document

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End of Section

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- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

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- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
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- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (262) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins. 14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service Representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

End of Section

REVISION HISTORY

REVISION	DATE	REASON FOR CHANGE
0	12/23/98	Initial Release.
1	11/9/99	Additional information regarding CT HiSpeed, minor corrections.
2	03/07/2002	LightSpeed information added, CT 9800 information added, Portable Building information added, minor corrections,

Mobile CT System Requirements Document

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Section - 1, General

1.1. Purpose of Document

This document is intended to identify requirements for the deployment of GE Medical Systems CT Scanners in mobile vehicles to ensure a safe and serviceable environment for personnel and equipment. Systems installed that do not meet the requirements of this document shall not be supported by GE Medical Systems. Warranty and service agreements may become voided if these requirements are not met.

GE Medical Systems reserves the right to refuse installation or delivery of any systems for a mobile, transportable, or re-locatable purpose. Exceptions or variances must be approved in writing by GE Medical Systems, CT Engineering Manager. These documents shall then be maintained for the life of the product.

This document is applicable to new CT systems specifically approved for installation in mobile environments. Please see addendum for approved CT models.

1.2. Intended Audience

This document is intended to provide guidance to vendors of new and used vehicles, sales personnel, installers, or engineers involved in the sales, placement, or installation decisions of GE Medical Systems CT scanners in a mobile, transportable, or re-locatable environment.

1.3. Document Precedence

Technical data service manuals, specific to various models of GE CT scanners, will take precedence over data herein, unless the data applies specifically to the mobile deployment of the scanner.

Other safety or general service policy directives will also take precedence, unless the data applies specifically to the mobile deployment of the scanner.

Section 2, Legal and Environmental Considerations

2.1. Legal

All installing vendors and customers purchasing a GE Medical Systems scanner to be installed in a mobile environment must assume responsibility for safe and legal employment or commissioning of the scanner. Disclaimers are made available for this purpose (see addendum) and must be signed by both the van manufacturer and the purchasing customer. This includes **compliance with all applicable** regulatory and law enforcement agencies including but not limited to the following:

NEC (National Electrical Code) standards

California Title 25

NCRP (National Council on Radiation Protection) standards

DOT (Department of Transportation) regulations

SAE (Society of Automotive Engineers) standards

TRA (Tire and Rim Association) standards

CFR 49 (Code of Federal Regulations-Transportation)

NFPA (National Fire Protection Association) regulations

ASHRAE (American Society of Heating, Refrigeration, and Air Conditioning Engineers) standards for H.V.A.C.

ANSI (American National Standards Institute) standards

TUV (Technischer Uberwachungs-Verein) (for European deployment)

Local country, state, or area laws and building codes

Local Health Regulatory Codes

OSHA

FCC

2.1.1. Electrical components used throughout the vehicle must be in compliance with certified listing agencies such as UL, ETL/CSA, or CE (for Europe).

2.2. Environmental. All customers purchasing a scanner to be installed in a mobile environment shall assume responsibility for maintaining environmental conditions of the scanner to avoid damage to the scanner or control systems.

A disclaimer is made available for this purpose (see addendum) and must be signed by either the van manufacturer or the purchasing customer. Failure to properly maintain equipment environment will result in voiding equipment warranties or service agreements.

2.2.1. When the equipment is operational, the environment **shall be maintained according to GE's published environmental specifications for equipment operations**. Each model has specific environmental specifications and these can be found in the Preinstallation manual for that scanner. (See addendum).

2.2.2. Appropriate air conditioning systems shall be installed and maintained for the environment the mobile units are to be operating in.

2.2.3. At a minimum, air conditioning systems shall be adequate to consistently maintain the van's internal environment to equipment specifications for all ambient external conditions from -20 degrees F through +110 degrees F @ 35% relative humidity and in direct sunlight.

2.2.4. When the equipment is not being operated (for any reason including storage or transportation), **the environment shall be maintained according to GE's published environmental specifications for equipment warehousing or storage**, found in the Preinstallation Manual for the specific scanner. For this purpose, appropriate air conditioning systems shall be installed, maintained, and powered in the mobile system to prevent damage to the equipment.

3. Power and Grounding Requirements

3.1. General. This specification covers electrical power and grounding requirements for protection from electrical shock and fire hazards.

3.1.a. Within the vehicle, the electrical power wiring shall be a 4 - wire (no neutral) grounded B-phase configuration.

3.1.b. The voltage of the input power to the CT must be 480 VAC for U.S. applications. For international systems, 400 VAC 50 and 60 hertz shall be used. Consult the CT Preinstallation manual for input configurations.

3.2. Equipment Grounding. The basis for the grounding requirements is to provide a “zero voltage reference”. The primary protection against electrical hazards is provided by the control of material used for conductors and insulators, the selection of components to meet power requirements, and the spacing of live electrical elements. Equipment grounding is intended to supplement this primary protection.

3.2.a. GE will provide grounding points on all of the CT equipment. This grounding point shall be attached to the CT’s main grounding point which is at the Power Distribution Unit (PDU) of the CT system.

3.2.b. This main grounding point shall be connected to the main input power ground connection with a copper wire.

3.2.c. An external earth ground lead (separate from the ground provided by the main shore power connection) should only be employed when the shore power connection is not possible and the vehicle will be occupied.

3.2.d. Should an external ground lead be necessary, the specified resistance of earth-to-ground is less than two ohms ($< 2 \text{ ohm}$).

3.3. Input power switch-box.

3.3.a The input power for the CT, shall have a main line power switch, mounted on a wall surface, unobstructed and within easy reach, located within 1.5 meters of the PDU.

3.3.b This power switch must be capable of being locked in the “Off” position using a standard padlock for Lock-Out-Tag-Out (LOTO).

3.3.c. The CT main input power lines shall also pass through a main power contactor, rated appropriately for the system being installed, and capable of being triggered by an Emergency Power Off (EPO) circuit. See the applicable pre-installation manual for breaker/fusing requirements. This contactor assembly may be integral to the switch-box or in a separate box in series with the switch box. This contactor shall be provide by GE with the CT system in some cases (A1), but if it is not provided, the installing van manufacturer shall provide it.

3.3.c.1. The CT main power lines must also pass through a shunt-trip type circuit breaker which shall be provided by the van manufacturer. This breaker shall be rated appropriately for the system being installed.

3.3.d. An under-voltage detection circuit for all three power phases shall also be installed to prevent powering up the system in the event of a break in any of the phases or in a low-voltage condition. When detecting a faulty power condition, the power monitor shall trip the shunt-trip breaker defined in the previous paragraph. The power monitor shall not be connected into the E-Off circuitry.

3.3.e. Paragraph deleted.

3.3.f. Schematic diagrams of the input power scheme shall be documented in the van maintenance manual , or separately if the CT is installed in a used van. Modifications to the van manufacturers original schematics may suffice for this purpose. All CT system components and options shall be shown.

3.4. Back-up power. All mobile CT systems shall have a back-up AC power generator to sustain the proper environment of the CT when shore power is not available. All vans shall meet the published vibration requirements with the generator running.

3.4.a. In general, it is not recommended to scan on back-up power as this may cause artifacts and needs to be grounded separately. If there is a need to scan on back-up power, the generator shall be minimally capable of 100 kilowatts constant output.

3.4.b. If the system does not require scanning on back-up power, the generator shall be minimally capable of 10 kilowatts continuous output power plus the power demand of the HVAC and other electrical equipment installed in the van. In this mode, the operator may have

the CT powered up and may perform other functions, such as filming, reformatting, or archiving while on back-up power.

3.5. Power switching.

3.5.a. Power source switching. Switching between shore-power and generator power shall be accomplished by an electronic switch. This switch may be located in the operator's room internally, or may be in the van's lower compartments. In either case, it must be clearly marked and easily accessible by the van operators.

3.5.b. Off Position. This switch shall have an "off" position which disconnects all power from all sources and shall employ "Break-before-make" contacts.

3.5.c. Switch Labeling. This switch shall be labeled for its intended purpose and it shall also have a label with the following text: "Caution, when switching between power sources, always stop in the 'Off' position for a minimum of five (5) seconds." These labels shall be provided by the installing van manufacturer.

3.5.d. Auxiliary contact. An auxiliary switch contact is required for high voltage lock-out, **if the back-up power generator is not sufficiently rated for scanning capability**. This additional contact shall be closed when in shore power mode and open in the back-up power mode.

3.5.e. If an auxiliary contact is required per the above paragraph, this circuit shall be wired in series with the scan-room door interlock circuit to prevent scanning while on generator power. Should the customer make a legal request to have the scan-room door interlock disabled, the bypass shall be accomplished at the scan-room door, not the GE equipment. In this manner, the scan-room interlock circuit will still be active for the purpose of preventing scanning on back-up power.

3.6. Shore power hook-up.

3.6.a. Systems shall use single cable (sometimes known as the "umbilical cord") to connect the van to shore power. Wiring shall be size 1/0 AWG (50mm²) for each of the three phases and also size 1/0 AWG (50mm²) for the ground wire.

3.6.b. The shore power connector shall be a 200 amp rated plug at a minimum. A 5-conductor, "RUSSELL STOLL" type plug is standard (See figure - 1). If another type of power connector is required, an adapter cable shall be provided.

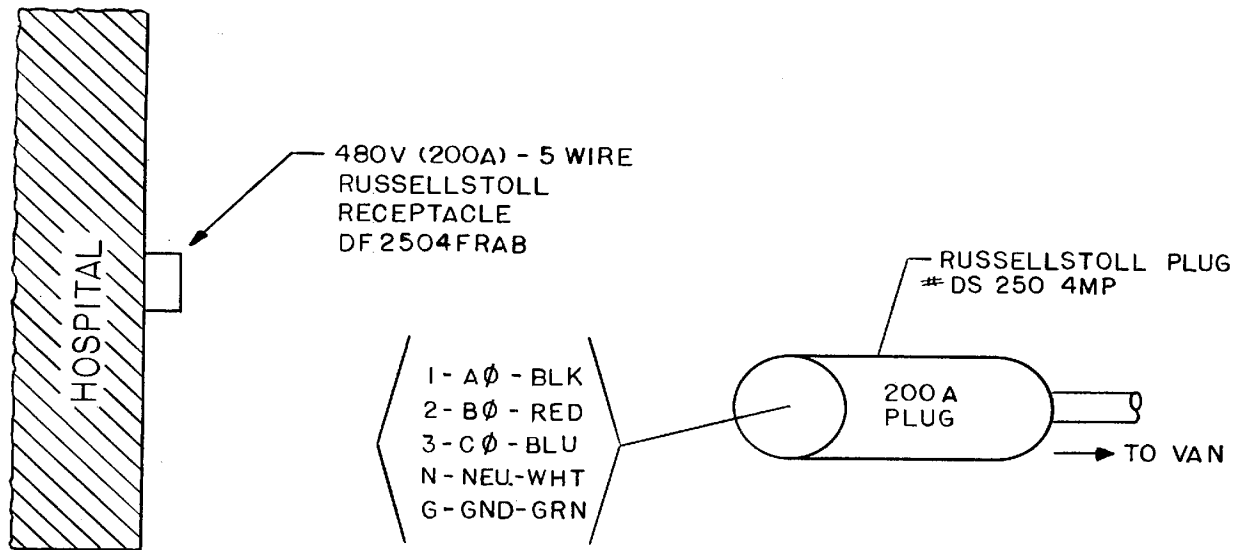


Figure - 1

3.6.c. Shore Power Step-up or Step-down transformer. If hospital power of the specified voltage for the mobile system is not available and a shore-power step-up or step-down transformer must be installed, the transformer shall have a constant output capacity of 150 KVA or higher. A "Delta to Y" secondary transformer is recommended.

Section 4, Suspension, Chassis, and Vibration Requirements

4.1. Suspension. All van or truck chassis to be used for mobile CTs shall have an air ride suspension on the rear bogies.

4.1.1. For vans (units which are pulled by a tractor), it is additionally required that the king-pin/bolster plate shall also have an air-ride suspension.

4.2. The suspension shall be in good working order. For used vehicles, the unit must be inspected by a certified truck/trailer service center prior to installing the CT in the van or truck. (D.O.T. Inspection).

4.3. All vehicles shall have an appropriate suspension for the estimated total gross vehicle weight requirement (GVWR) which will be transported plus any additional weight added when parked or scanning. Total weight, distributed weight, and axle weight for any axle, must conform to DOT guidelines.

4.4. All vehicles or vans shall have a GVWR chassis of at least 40,000 pounds. Variations may be considered but must be specifically approved by GE CT Engineering.

4.5. All vehicles (vans and self contained vehicles) will need to meet this shock and vibration environmental specification. This environment will take into account normal transportation of the mobile system over primary and secondary roads within the United States, Europe and Japan.

RANDOM VIBRATION:

Frequency Hz	PSD Level (g^2/Hz)	Slope (dB/octave)
5 - 20	0.00025	
20 - 30		-50
30 - 50	0.00001	
Total rms = 0.055 g _{rms}		

Total g_{rms} level is not to exceed 0.055 g_{rms} is based on a typical spectrum meeting the above specified rms levels, not the total energy under the specification curve.

Measurements are to be made over the rear suspension, which is typically the base of GEMS equipment, and over the fifth wheel of the trailer or the front of a self-container vehicle.

SHOCK:

- Data low pass filtered at 100Hz.
- 1.5g Vertical and Transverse
- 1.2g Longitudinal

4.5.1. Used vehicles manufactured by Calumet Coach Company, Ellis and Watts, A.K Associates, Synergetics, Smit, or MDG that have been certified previously, have been tested and are assumed to already meet the above specification. Prototype vehicles and used vehicles that are from a manufacturer not listed above, shall be road-tested by GE's Global Mechanical Technology Center (GMTC) to insure that they meet the above specifications.

There may be other vibration considerations depending on the model of CT to be installed. See addendum for the specific scanner.

4.6. Additional vehicle requirements.

4.6.1 A red transit warning lamp shall be illuminated if the CT system is not prepared to transport. (Not applicable to CT 9800 mobiles). On trailer type units, this lamp shall be mounted on the driver's side of the unit, visible in the rear-view mirror. On self-propelled units, this lamp shall be located on the driver's dash-board. This lamp shall be provided by the installing vendor.

4.6.2. GE will use two or more normally closed contacts to indicate that it is not prepared for transportation. When any of these contacts are closed, the warning lamp must be illuminated to warn the driver. Specific interface information is contained in the service manuals for each scanner.

4.6.3. This warning lamp shall be on a 12 volt DC, battery powered circuit utilizing two bulbs. A charger for the battery shall be provided by the installing vendor.

4.6.4. The warning lamp shall be labeled **"Transit Warning - Caution: Do not move this vehicle if this lamp is on. Check unsecured items"**.

4.6.5. The circuit may also be used for other non-CT preparations. (Such as interlock switch if the patient-lift must be stowed prior to transit).

4.6.6. Documentation to instruct transit preparation shall be provided with the vehicle. This shall be provided in this documentation and shall also be posted on the scan-room wall: See addendum of specific CT for transit preparation instructions. It is the responsibility of the installing vendor to post these instructions on the scan-room wall.

4.7 Shore-Station Filming. For safety reasons, if a connection is required for shore-station filming, this connection shall be of an optical or other wireless configuration.

Section - 5, Equipment Mounting Requirements

- 5.1. General. All equipment shall be mounted or secured in a safe manner.
- 5.2. All equipment shall be secured to the floor or walls and be capable of preventing movement or shifting in the event of a panic stop.
- 5.3. All equipment covers shall remain in place. Equipment shall be positioned in such a way to facilitate removing and reinstalling equipment covers for normal service.
- 5.4. No modifications to the GE equipment or covers is allowed without specific, written, approval by GE Global CT Engineering.
- 5.5. GE Equipment identification or rating plates shall not be covered and shall be visible without moving the equipment.
- 5.6. Scanner components. An addendum is provided for each type of scanner regarding specific equipment mounting requirements. See addendum.
- 5.7. Optional components. All electronic equipment (including cameras, networking devices, interface boxes, etc.), shall be mounted using metal brackets, straps, and “D-Rings” to prevent shifting or movement in the event of a panic-stop. Velcro alone or “bunji cords” are not sufficient. These restraints will be provided by the installing vendor.
- 5.8. Straps or retainers shall not require tools to remove.
- 5.9. Van manufacturers shall have documented Engineering evaluations indicating that all straps and retainers are sufficient for non-CT system options.
- 5.10. Steel mounting plates underneath each component shall be flush with the floor or have beveled edges so that equipment may be rolled on and of plates for installation or servicing.

Section - 6, Safety and Serviceability Requirements

6.1. General. All mobile environments shall provide adequate access to the GE equipment in order to allow for safe, effective operation and servicing of the equipment.

6.1.1. Equipment shall be positioned in the vehicle in a manner which provides appropriate service access. For this reason, proposed layout diagrams must be provided to GEMS CT Engineering for preliminary review prior to the start of any mobile construction or equipment installation (unless it is a pre-certified configuration).

6.1.2. The diagrams must show all interior room dimensions, service door or slide-out dimensions, and clearances provided around all equipment. All other non-CT equipment or cabinets intended to be installed must also be shown so that possible interference may be analyzed.

6.1.3. All signal and cable ducts for system interconnection shall not interfere with GE equipment or servicing. I.E., no above-the-floor ducting is allowed if it will hinder service or operations in any way.

6.2. Equipment placement in the vehicle must allow for the following:

6.2.a. There shall be adequate room to remove all covers. If there is insufficient room to remove any of the CT system covers in the usual manner (by using the normal service tools provided with each system), alternative means such as slings, tracks, and hoists may be designed and installed by the van company doing the CT installation. In this case, GE CT Engineering must approve the alternative means. Additionally, this alternative method shall be tested and instructions shall be documented and posted by the installing van company.

6.2.b. Equipment covers shall not be modified in any way without the specific written approval of the GE CT Engineering Manager.

6.2.c. No "single-person" task shall become to a "two-person" task.

6.2.d. If a service hatch or expansion area is required in order to access or remove covers, the expansion area must maintain a closed environment. Open clamshell platforms or service doors shall not be used. Slide-outs, fold-outs, "pop-outs" are all allowed providing that they maintain the environment. Exception: GE CT Mobile installations completed prior to January 2000, may have clam-shell type service doors.

6.2.e. Canvas coverings of service hatches which must be snapped or hung in place after opening the area are not a suitable method of maintaining the environment and shall not be used. A service only hatch that has expandable canvas sides which maintain an air seal may suffice.

6.2.f. No external access via ladders or outside steps should be needed. Adequate clearance shall be available for inside access to the service hatches. External steps may also be supplied for loading equipment or parts, but are not intended to be the only route of access. In other words, the service personnel should not need to exit the vehicle and walk around to another entrance in order to access the equipment for normal service. Latches or locks that must be operated outside the van in order to expand service hatches are acceptable only if they can be operated while standing on the ground.

6.3. Reviews. All used vans and all prototype or uncertified new vans, shall have safety and serviceability inspections/reviews by GE qualified personnel after the equipment is installed in the vehicle. If items noted on these reviews are not corrected or resolved, GE shall refuse to deliver the system, or may void the warranty or service agreements.

As a general reference of what these reviews consist of, see the sections for new vehicle (Section - 7) or used vehicle (Section - 8) considerations for sample inspection sheets.

6.4. Lead shielding. All CT Mobile installations occurring after 1 January 2002 shall contain the following level of lead shielding at a minimum:

- 1/16 inch thick lead lining (or equivalent) of all four scan-room walls, from floor to ceiling, including doors, slide-outs, and windows.
- 1/16 inch thick lead lining (or equivalent) in the slide-out floors and in the scan-room floor perimeter. This floor perimeter area lead shielding shall be adequate to prevent any straight-line exposure from the CT gantry aperture iso-center to the ground directly outside of the van.

Section 7, New Vehicle Considerations

7.1. General. Should a vendor desire to install and sell mobile GE CT systems, they must work with GEMS CT Engineering to develop a configuration that is agreed upon by both parties. The agreement will include the design specifications of the vehicle and will also include the scope of work involved for both parties during manufacturing and installation.

Upon agreement of a suitable, safe, and serviceable mobile configuration, the vendor may seek design certification for that configuration (see process identified below). A design certification letter will be issued to the vendor at that time.

Once a configuration is certified, only changes to the configuration need to be approved for future sales of subsequent systems. Minimal interaction will be required once a configuration is certified. Desired changes to a certified configuration must be submitted via Van Change Request (VCR form which is provided by GE) for approval by GEMS CT Engineering prior to implementing the change.

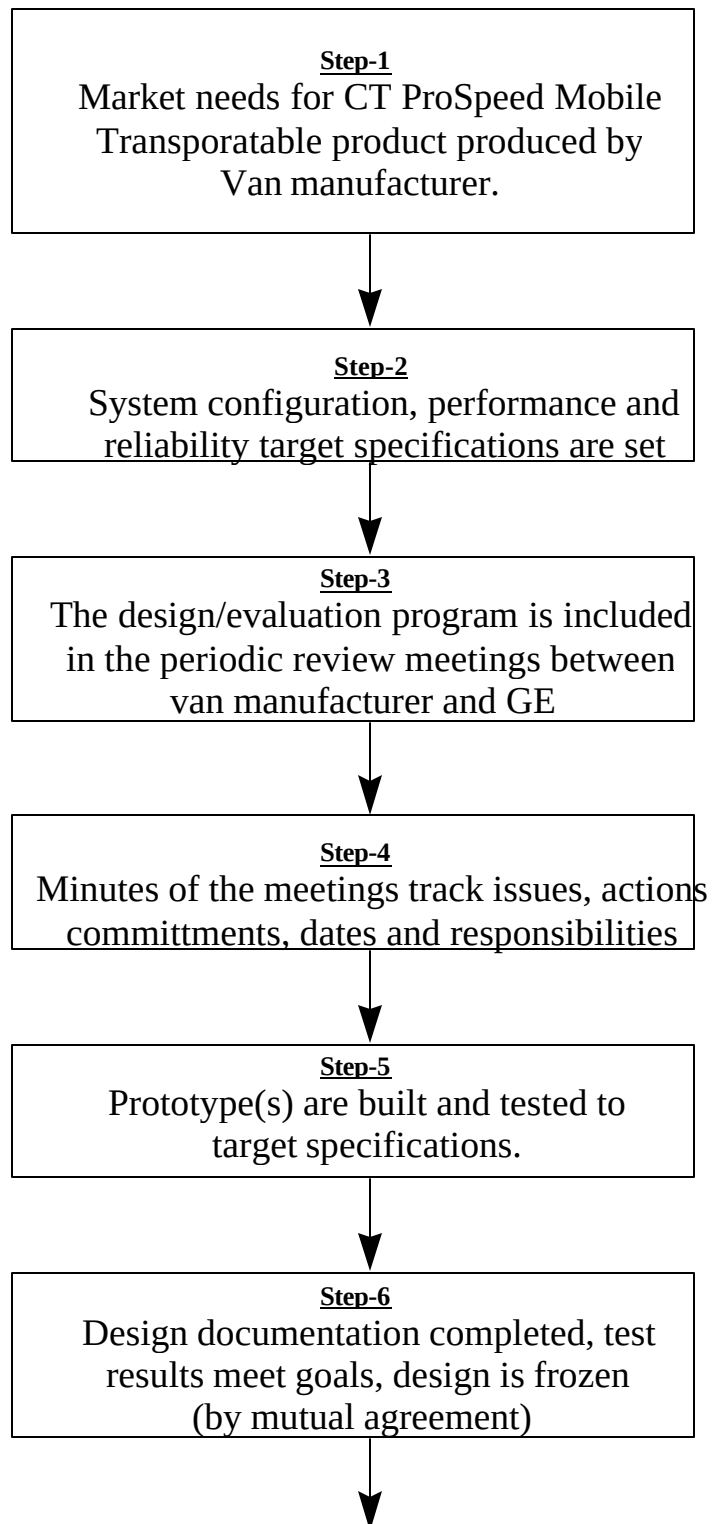
GE reserves the right to revoke certification at any time.

7.2. Documentation Requirements. The following documents shall be under revision control and shall be provided to GE CT Engineering prior to certification:

- Van Drawings (including structural, electrical, and proposed systems layouts)
- Van Service Manual
- Van Operator's Manual
- Site Planning Guide
- Design Compliance Checklist (Format available from GE)
- Design Certification Checklist of prototype (Format available from GE)
- Road Test Results (Done by GE's GMTC Group)
- Serviceability Review (Final Review will be done by GE after initial installation)
- Van Change Requests (VCRs)
- VCR Matrix (When applicable)

7.3. Certification Process. The following flow-chart describes the basics of the certification process:

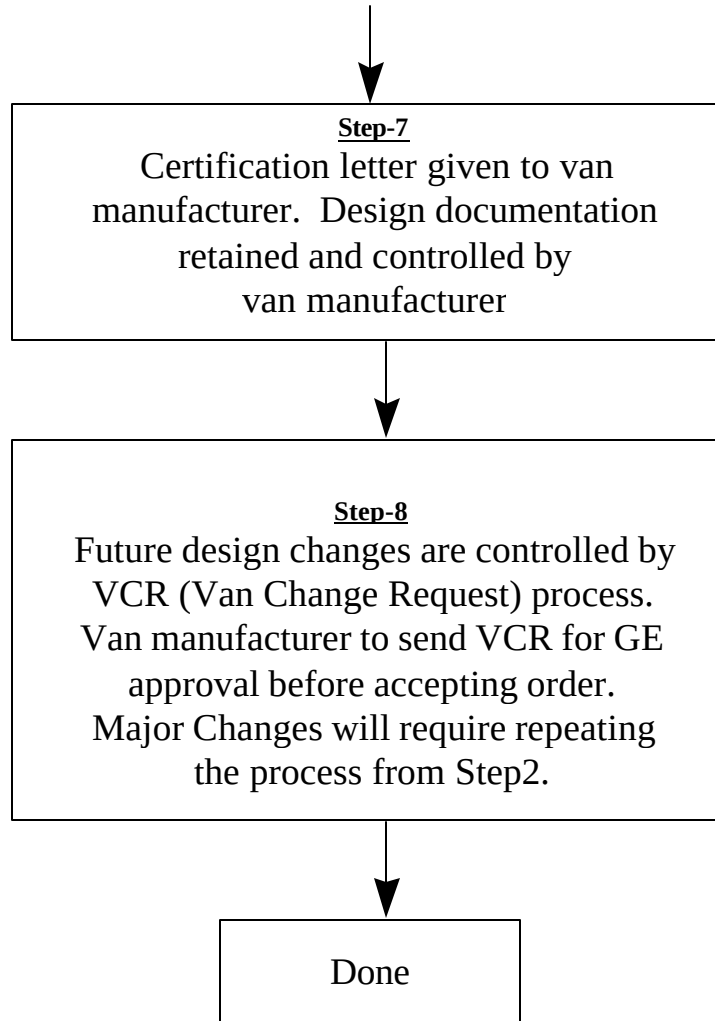
CT MOBILE DESIGN/EVALUATION and CERTIFICATION
PROCESS FLOW CHART



Go to next page:

CT MOBILE DESIGN/EVALUATION and CERTIFICATION
PROCESS FLOW CHART (continued)

From previous page:



8. Used Vehicle Considerations and Documentation Requirements

8.1. General. GEMS CT Systems may be installed in a used vehicle or van, only when the van meets specific design and serviceability requirements. Checklists are made available herein to insure that the mobile vehicle meets the requirements.

8.1.1 The physical installation of the CT into the vehicle shall be done by a qualified vendor in order to use the expertise needed to safely mount the system, insure that the vehicle complies with all applicable rules and regulations, and to insure that the subsystems are working properly. The vendor shall be qualified by GEMS CT Engineering.

8.2. Process. See the following Step-by-step process:

Step 1: GE Sales Representative identifies an opportunity to replace a scanner in an existing mobile environment. All used vehicle prospects shall be a three-party project (The customer or end-user, GE as a provider of the CT, and a qualified mobile installation vendor to perform van preparation for the CT, CT installation, and any other van reconditioning that the customer may desire). A Phase-1 (Preliminary) checklist is performed to determine if vehicle is suitable.

Step 2: If the vehicle passes the preliminary inspection, a qualified mobile installation vendor must be identified for van preparation, installation of the CT, and any other modifications required by the customer. Contact CT Engineering to identify qualified vendors.

Step 3: The qualified vendor inspects the van to determine changes required and to determine if the van sub-systems are adequate. A preliminary quote may be made, reserving costs for unforeseen changes that may be required. A Phase-2 checklist is performed by the vendor.

Step 4: If the van is still considered suitable, a statement must be obtained from the van vendor, that all of the van subsystems are adequate or can be made adequate. These subsystems include:

- The air conditioning and heating systems including ventilation (See section 2).
- Electrical Power and Grounding system (See section 3).
- Back-up power generator (See section 3).
- Suspension and DOT compliance (See section 4).
- Radiation Shielding if additional shielding is needed per paragraph 6.4.

Step 5: The vendor shall provide GEMS CT Engineering with a proposed drawing of the vehicle layout and upgrade procedure. (See section 6.1).

Step 6: GEMS CT Engineering will review the proposed layout and approve or propose changes. If the requested changes are incorporated, approval may be obtained.

Step 7: Once the configuration and all associated costs are approved, the installation may proceed. The vendor must accomplish the Phase-3 (Installation) checklist. Disclaimers must be signed by both the customer and the vendor. A GEMS CT Design Compliance Checklist shall be completed by the van manufacturer and forwarded to the GEMS Global Mobile Manager.

Step 8: After CT system installation, the van will be transported to the GEMS Mobile Facility (in Waukesha Wisconsin) for final calibration and inspection by the GEMS Mobile Manufacturing team. Discrepancies shall be identified and corrected. Van company issues will be identified for correction. If it is not feasible to make immediate corrections, a plan for correction shall be created.

Step 10: The system is returned to the installing van manufacturer for correction of any remaining issues, van clean-up, and customer delivery.

8.2.a. Exceptions. Systems being installed outside the United States shall have local GEMS personnel perform final inspection and calibrations. Any other deviations from the above mentioned steps must be approved by GEMS CT Engineering.

8.3 Documentation Requirements. All modifications of the original van design shall be fully documented and provided to GEMS CT Engineering as well as leaving a copy in the van for service personnel.

Changes to the original van documentation may be made or separate documentation may be used for this purpose.

At a minimum, the following shall be provided:

- Schematics of the equipment layout.
- Schematics of the electrical system including locations of all components and van interface.
- Procedures for hooking up power and for switching power.
- Transport Preparations and siting instructions.
- Changes to the radiation shielding and Radiation Survey (when performed).
- Instructions to open or secure service entrances (These may be labeled directly on the entrances).
- Operating instructions for any special service tools.

8.4 All GE provided service documentation shall remain in the vehicle.

9. Portable Buildings

9.1. General. This section applies to all portable buildings including modular buildings, relocatable, and ISO containers which may be transported as a whole unit. If the building must dismantle prior to transport, the CT system must be removed prior to transport. In this case, the system is not considered to be a mobile unit.

9.2. GEMS CT Systems may be installed and transported in a transportable building, only if the CT scanner is a Mobile Approved version or if the shelter will not move with the CT system installed. Installation of CT systems in portable buildings whereas the building will be transported with the CT system installed in the building, must be installed by approved vendors only.

9.3. Prior to installation of a CT in a transportable building, GEMS Global CT Engineering must be provided with room layouts, system wiring diagrams, and HVAC design parameters in order to assess performance and serviceability.

9.4. All design requirements such as shielding, environmental, serviceability, hardening, securement, and documentation must be in compliance with the requirements set forth in this document with the following exceptions:

9.4.1. Monitors, keyboards, and other loose items which would require securement for transport may be removed and boxed for transportation in lieu of being strapped down.

9.4.2. No "Transit Warning" circuit is required.

9.4.3. Complete recalibration and system audit must be accomplished after transport and siting at customer expense.

9.4.4. A back-up power A/C generator is not required for transport. However, be aware that some components may have environmental limitations that need to be considered. Failure to protect the components from freezing, overheating, or condensation may result in additional charges for repair or replacement.

9.5. Transportable buildings loaded with GE scanners must be transported on an Air-ride suspension truck or trailer.

9.5.1. If the transportable building is meant to be transported by any other means (such as steel spring trucks, helicopter, ship, bogies, etc.), the CT system may need isolation, stabilization, or shock absorbing equipment. GEMS Global CT Engineering and the GEMS Global Mechanical Technology center must approve this design prior to installation.

A.1. Addendum of Disclaimers

A.1.1. General. The following pages contain legally approved disclaimers to be included as documentation that must be signed by any vendors or customers purchasing CT systems for use in a mobile environment.

Select the appropriate disclaimer appropriate for either a customer (A1.2) or vendor (A1.3). Make a copy of this disclaimer on GE paper and have a representative of the customer or vendor company sign the document confirming their understanding.

A Copy of signed disclaimers should be sent to :

GE Medical Systems, CT Engineering
Attn: Mobile Product Team Leader
Mail Station NB-918
Box 414
Milwaukee WI 53201

A1.2., Customer Disclaimer and Memorandum of Understanding

All customers purchasing a scanner to be installed in a mobile environment must assume responsibility for safe and legal employment or commissioning of the scanner. This disclaimer must be signed by a responsible individual of your company.

As a representative of _____, I hereby state that I understand that this company assumes responsibility for safe and legal commissioning of this scanner. This includes compliance with all applicable country, state, and local regulatory and law enforcement requirements.

It is further understood, that the environment for this scanner, shall be maintained according to the published specifications (found in the Preinstallation manual included with the CT scanner or purchased separately from GE Medical Systems) and compliance to all van siting requirements.

Signature: _____ Printed Name: _____

Title: _____ Date: _____

Van Model and Serial Number: _____

A1.3. Vendor Disclaimer and Memorandum of Understanding

All vendors servicing or installing GE Medical Systems scanners in a mobile environment must assume responsibility for safe and legal employment or commissioning of the scanner. This disclaimer must be signed by a responsible individual of your company.

As a representative of _____, I hereby state that I understand that this company assumes responsibility for:

1. Safe and legal commissioning, installation, or service of this scanner.
2. Compliance to the requirements set forth in the GE Medical Systems Mobile CT System Requirements Document, GE part number 2218879-100.
3. Compliance with all regulatory and law enforcement agencies including but not limited to the following, **whenever applicable to the work being performed by our company:**

NEC (National Electrical Code) standards

California Title 25

NCRP (National Council on Radiation Protection) standards

DOT (Department of Transportation) regulations

SAE (Society of Automotive Engineers) standards

TRA (Tire and Rim Association) standards

CFR 49 (Code of Federal Regulations-Transportation)

NFPA (National Fire Protection Association) regulations

ASHRAE (American Society of Heating, Refrigeration, and Air Conditioning Engineers) standards for H.V.A.C.

ANSI (American National Standards Institute) standards

TUV (Technischer Uberwachungs-Verein) (for European deployment)

Local country, state, or area laws and building codes

Local Health Regulatory Codes

OSHA

FCC

It is additionally understood that the environment for these scanners, shall be maintained according to the published specifications while it is in our possession. (Specifications can be found in the Preinstallation Manual included with the CT scanner or purchased separately from GE Medical Systems).

Signature: _____ Printed Name: _____

Title: _____ Date: _____

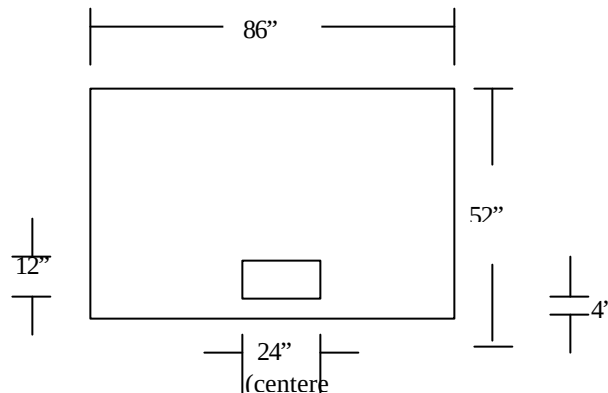
A.2. Addendum for all Prospeed Systems

A.2.1. General. Prospeed systems must be of a valid configuration for mobile considerations. Not all of the Prospeed line of scanners are suitable for mobile applications. In general, all ProSpeed series scanners manufactured after November 1997 are considered suitable. For further assistance in identifying these scanners, contact GEMS Global CT Engineering.

A.2.2. Preliminary planning. **All vendors intending to install a Prospeed scanner should first consult the Prospeed Preinstallation Manual, part number 2124010 and the Prospeed Mobile Maintenance Manual, part number 2182577** This manual includes all technical specifications pertaining to the siting of the GE Prospeed scanners including equipment weights, desired clearances, and environmental conditions. **These documents can be ordered by contacting the GEMS Direct Customer Order Service group at 1-800-558-2040.**

A.2.3. Gantry Mounting considerations:

A.2.3.1. The gantry shall be mounted onto a steel plate. This plate shall be at least 3/4 inch thick. The steel plate will be 86" x 52", with a 12" x 24" rectangular cut-out centered 4" from the rear edge of the plate (for cable routing from below deck). See the drawing that follows:



A2.3.2. The steel gantry mounting plate shall be appropriately welded using gussets to the main chassis or frame members of the truck or trailer. **Note: This plate is the main support platform for the entire weight of the gantry and must be able to withstand all the stress of a panic stop in addition to normal tilt and scan operations. You may need to consult a structural engineer for actual loading calculations.**

The mounting plate may be mounted below the surface level of the floor to allow for seamless tiling.

A2.3.3. All ProSpeed gantries arrive with four angle brackets welded onto the ends of the gantry base. Two holes shall be drilled into each of these angle brackets for the mounting bolts. (A total of eight mounting bolts will be used to secure the gantry to the steel mounting plate).

A2.3.4. There will already be at least one hole in each of the four angle brackets which is pre-drilled and tapped for leveling feet. The leveling feet shall be removed for mobile applications and each of the threaded holes can be re-drilled and used as a pilot hole for one of eight mounting bolt holes.

A2.3.5. Eight (8) steel bolts shall be used for mounting the gantry to the steel plate. Use 5/8 inch or M16, type A449 high-strength bolts or equivalent. Grade 5 bolts are also suitable. Tighten to proper torque specs.

A2.3.6. The steel plate shall be drilled and tapped for the eight mounting bolts. This may be done after the gantry is physically positioned on the steel mounting plate in order to have exact placement for drilling (Using the holes in the angle brackets as a template).

A2.3.7. Spacers shall be put between the lower surface of the angle brackets and the upper surface of the steel plate. Aluminum or other suitable material may be used for this purpose. See illustration below.



A2.4. Table Mounting Requirements.

A2.4.1. The table shall be mounted onto a steel plate which may be built into the floor to allow for seamless tiling. The steel plate shall be 60" x 24" (minimum) and shall be at least 1/4 inch thick. The steel mounting plate shall be secured to the frame of the vehicle. **Note: This plate is the main support platform for the entire weight of the table and patients. It must be able to withstand all the stress of a panic stop in addition to normal patient load and**

scan operations. You may need to consult a structural engineer for actual loading calculations.

A2.4.2. After GE installation personnel have aligned the table to the gantry, four ½ inch holes must be drilled and tapped for securing the table. Four ½ inch hi-strength bolts shall be used for this purpose.

A2.5. Operator Console Mounting Requirements. The Operator Console shall be mounted directly in front of the scan room window, facing the front of the vehicle, approximately 12” from the Scan room wall. (This allows for adequate cabling room behind the O.C.)

A2.5.1. A 2”x 2”x 24” steel bracket shall be used to anchor the front of the O.C. to the floor using lag bolts. Existing screw holes in the base of the O.C. should be used to anchor onto the O.C base. See illustration following.



Figure A2.5.1. O.C. Front mounting bracket

A2.5.2. Additionally, the O.C. shall be anchored to the scan room wall. This is accomplished by a cinch strap going around the O.C. (just below the keyboard) to prevent tipping in the event of a panic stop. Two D-Rings must be attached to the wall for anchoring the cinch strap. The D-Rings should be approx. 30 ½ inches above the floor. (See illustration below).

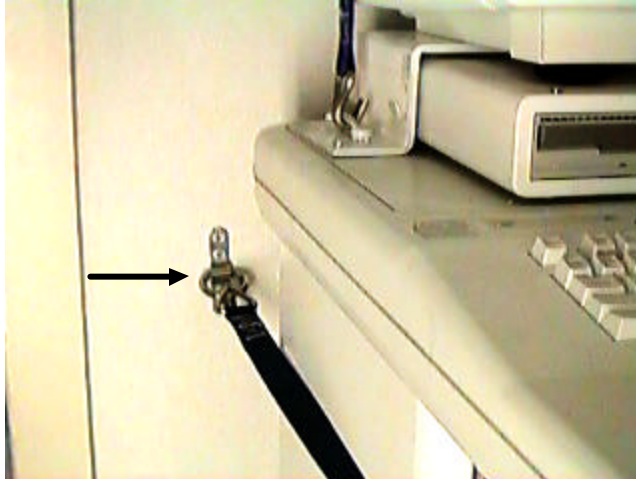


Figure A2.5.2. O.C. Wall Mount (Note: the cinch-strap seen above goes between D-Rings on both sides of the O.C. Right side D-Ring is not shown.)

A2.5.3 CRT Mounting Bracket. A steel plate shall be formed for mounting the CRT to the O.C. with a cinch strap. This steel bracket shall be formed so that it will cover the MOD and provide an anchoring mechanism for the CRT. The exact dimensions of the MOD are: $2 \frac{3}{4}$ inches high x $13 \frac{1}{8}$ inches wide. Thus, to provide necessary clearance, size the bracket for an inside dimension of $2 \frac{7}{8}$ inches x $13 \frac{3}{8}$ inches. See illustration below.



Figure A2.5.3. CRT/MOD Mounting Bracket

A2.5.4. Process. The vendor shall use the following basic process to mount the CRT:

- Step 1. Manufacture and paint the mounting bracket.
- Step 2. Remove the two rear feet of the CRT.
- Step 3. Attach the CRT to the mounting plate (where the rear feet were removed) with two screws.
- Step 4. Attach the mounting plate and strapping rings to the top cover of the console.
- Step 5. Attach a cinch strap

A2.5.5. Power Distribution Unit (PDU). The PDU shall be anchored to both the wall and the floor using an approved dampening system.

A2.5.5.1. The dampening system consists of the following:

- A steel bracket to be attached to the top of the PDU and the wall. (Supplied by vendor.)

- Two Barry Mount # 21335-5 stabilizers* for attaching the steel bracket to the wall.
(Supplied by vendor.)
- Four SLM-3 air cushion isolators* for attaching to the floor. (Supplied by vendor.)
- A hardening kit for the internal circuit board. (Provided in the GE Mobile Installation Kit.)

*Contact GEMS CT Engineering if a purchasing source is needed for these items.

See illustrations below:



Figure A2.5.5.1. PDU Wall-Mount Bracket with Barry Mount Stabilizers.



Figure A2.5.5.2. PDU Mounted on Barry Mount Isolators

A2.6. Table Transport Bracket. (Commonly known as the “Ski” bracket). A transport stabilizing bracket shall be configured. This bracket consists of the following:

1. A base-plate & tongue assembly (Provided in the GE Mobile Kit).
2. A cross-bar (Provided in the GE Mobile Kit).
3. Two D-Rings recessed into the floor near the table (Supplied by vendor). These D-Rings shall be located on the floor, 28 inches from the table base, and 35 inches from the gantry base.
4. Two 1-Inch Cinch-Straps (Supplied by vendor).

See the following illustrations: (On next page)



Figure A2.6.1 Table Transport Bracket (Mounted into end of cradle).

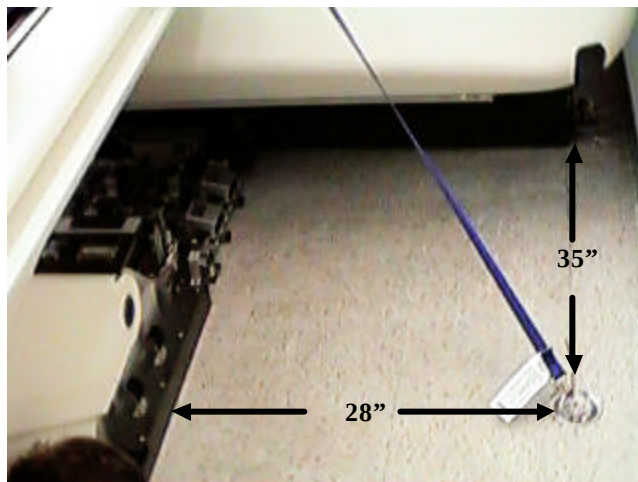


Figure A2.6.2. Table Transport ("Ski") Bracket Cinch-Strap and D-Ring, Right Side
(Left side not shown).

A2.6.1. Process. The following installation process shall be used to configure the Ski Bracket:

Step 1. Assemble the base plate to the cross-bar.

Step 2. Mount the D-Rings into the floor.

Step 3. Attach the cinch-straps to the bracket.

Step 4. Attach caution label to the bracket. Caution label should state:

“Caution: This bracket is for transport only.
Install prior to transporting vehicle and remove
prior to applying CT power.”

A2.7. Instructions for Transit Preparation shall be mounted on the wall of the scan room. See next page for these instructions.

GE CT ProSpeed Mobile Transport Preparation

The following procedure must be accomplished prior to transporting this vehicle: If these items are not accomplished, the “Transit Warning Lamp” will remain illuminated on the outside of the van/truck.

Step 1: Retract the cradle completely (out of the gantry) and remove all accessories.

Step 2: Raise the table to it's highest limit.

Step 3: Tilt the gantry to it's backward limit.

Step 4: Lower the table until it stops.

Step 5: Install the transport bracket into the end of the cradle.

Step 6: Use cinch-straps to secure the ends of the bracket to the “D-Rings” in the floor near the table base. Tighten the cinch straps just to the point that the straps are straight and snug. Caution: do not over-tighten.

Step 7: Turn the system power off at the Operator's Console.

Step 8: If the system will not be on back-up power during transportation, turn power off at the main disconnect.

Step 9: Perform any other transport preparations required per the van Operator's Guide or other applicable documentation.

A3 Addendum for all HiSpeed or “Xi” Mobile Systems

A3.1. General. HiSpeeds systems must be of a valid configuration for mobile considerations. “Mobile” versions of the “GE HiSpeed CT” scanners are hardened at the factory to be suitable for mobile applications. For further assistance in identifying these scanners, contact GEMS Global CT Engineering.

A3.2. Preinstallation and Installation information. Required information for the installation of HiSpeed Mobile CT systems is provided in the standard CT HiSpeed Preinstallation and Installation manuals. These manuals are GE part numbers 2188540 (Rev 8 or higher) and 2202122 (Rev 12 or higher) respectively. These manuals arrive with the system. **Additional copies can be ordered by contacting the GEMS Direct Customer Order Service group at 1-800-558-2040.**

A3.3. Mounting Requirements. All mounting hardware is provided with the HiSpeed Mobile CT system. However, the mounting bolts shall have threaded placements within the van for securing components onto. These threaded locations shall have at least a 1 to 1 ratio of steel thread compared to the diameter of the mounting bolts. I.E., a 12 mm bolt must have 12 mm or more of threaded steel to adhere to. This steel shall be appropriately engineered and mounted to the frame of the van to withstand repeated loading and panic stops. Steel plates or other effective means may be used. Component weights are noted in the Preinstallation Manual. Additionally noted in the Preinstallation Manual is a “flatness” specification ($\leq 2\text{mm}$) for all component mounting areas which shall be met.

A3.3.1. Gantry and Table. A $\frac{3}{4}$ inch steel plate shall be provided by the van manufacturer for both the gantry and the table. These plates shall be appropriately engineered and mounted to the frame of the van to withstand repeated loading and panic stops. A diagram including mounting bolt locations is provided in the PreInstallation Manual.

A3.3.2. Operator’s console and PDU. $\frac{1}{2}$ ” or 12 mm steel plates in the floor are recommended. Alternative means may be used if they comply with paragraph A3.3.

A3.4. Transit warning system interface. An interface for the transit warning system is provided in the base of the Operator’s Console. The CT HiSpeed system has a circuit of interlock switches to provide a warning when the CT system is not prepared for transport. No power is provided by the CT system. The switches are “closed” when the system is not prepared for transport. See the Installation Manual for connection details.

A3.5. “X-Ray On” lamp interface. An electrical interface for “X-Ray On” lamps is provided in the base of the gantry. See the Installation Manual for connection details.

A3.6. Scan Room Door Interlock. An electrical interface for the scan room door interlock is provided in the base of the gantry. See the Installation Manual for connection details.

A3.7 CT Transport Preparation. The following page contains a procedure which needs to be accomplished by the operators or truck driver in order to prepare the CT for transportation. This procedure shall be mounted on the interior wall of the scan room by the installing vendor.

A3.8 EKG/Cardiac Scoring Option. Should the customer require the Cardiac Scoring option for the CT HiSpeed, an EKG monitor shall be mounted on the road-side scan-room wall near the table. A bracket will be provided for this by GE. The customer will use leads that connect between the EKG monitor and the patient. A signal cable will connect from the rear of the EKG monitor to the rear of the gantry.

Van companies are responsible for mounting the EKG monitor, providing power to the unit, cabling the EKG to the gantry, and designing a transport securing method. Design for the mounting, cable run, and transport securing must be approved by the GEMS CT Engineering Mobile Product Manager.

CT HiSpeed Transport Preparation

The following procedure must be accomplished prior to transporting this vehicle. If these items are not accomplished, the “Transit Warning Lamp” will remain illuminated on the outside (driver’s side) of the van.

Step 1: Retract the cradle/table top completely (out of the gantry) and remove all accessories. If the CT system is equipped with a moving cradle base (IMS assy), make sure it is also retracted.

Step 2: Raise the table to it’s highest limit.

Step 3: Tilt the gantry to it’s backward limit.

Step 4: Perform the “Shut-down” procedure at the Operator’s console and then turn off the system power (at the front of the Operator’s Console).

Step 5: Use two cinch-straps to secure the steel rings underneath the front of the table to the “D-Rings” in the floor near the table base. Tighten the cinch straps just to the point that the straps are straight and snug. Caution: do not over-tighten.

Step 6: Install the transport bracket into the end of the cradle/table-top.

Step 7: Use two more cinch-straps to secure the ends of the transport bracket to the same “D-Rings” in the floor near the table base. Again, tighten the cinch straps just to the point that the straps are straight and snug. Caution: do not over-tighten.

Step 8: Pull up the two red handles at the base of the Operator’s Console.

Step 9: If your system has the Cardiac Scoring option (with EKG monitor), secure the EKG monitor into the transport position with the straps provided.

Step 10: The CT system is now ready to transport. Accomplish all other van preparations according to the van manufacturer’s instructions.

Set-Up Instructions

To prepare the CT for scanning, simply reverse the above procedure, undoing steps 9, 8, 7, 6, and 5. Then power up the system by pushing on the main power button on the front of the Operator's Console. The system will boot up automatically

A4 Addendum for all LightSpeed Mobile Systems

A4.1. General. As of this time, only GE LightSpeed CTs that are ordered new as a Mobile unit from the factory are considered eligible for installation in a mobile environment. And, the mobile LightSpeed can only be delivered for installation to GEMS Certified van partners/manufacturers.

A4.2. Van Requirements. Only extra-wide vans (102 inches for U.S., 2200 mm for Europe) with dual slide-outs are eligible for CT LightSpeed installations. This extra width is needed because of the gantry width. Also, as with all CT Mobile systems, GEMS CT Engineering must certify any van for CT LightSpeed installations and will only work certified van partners to do so. GEMS CT Engineering will work with these partners to insure safety, serviceability, and performance.

A4.3. Documentation. In addition to this document, the following documents apply to CT LightSpeed Mobile scanners. Most of the special requirements and processes pertaining to CT LightSpeed Mobile will be found in these documents:

- Operator's supplement: 2321165-100
- Transit Prep statement posted on wall.
- On-Site Set-up: 2318669-100
- LightSpeed Mobile Specs: 2305587-100
- LightSpeed Mobile Mech Install: 2305588-100
- LightSpeed Mobile Elec Install: 2318573-100
- UPS Installation: 2315243-100
- Power-Up sequence posted on A-1
- I-Serve: 2320359

A4.4 Transit Warning Lamp. As with all current CT Mobile systems, a Transit Warning Lamp is required on the outside, driver's side of the van. (See section 4.6.) In the case of LightSpeed systems, the only switch for this lamp is a micro-switch mounted in the CT gantry, on the STC side tilt-stop block. If the gantry is tilted completely backwards (for transportation), this switch will open, extinguishing the lamp outside. If the gantry is not completely tilted backwards, this switch is closed and the lamp outside will be illuminated to warn the driver not to move the van.

A4.5. Transit Preparation and Set-Up Statements. The following two pages contain the text of the CT LightSpeed Transport Preparation instructions and a Set-Up Instruction which must be posted on the scan-room wall within each LightSpeed Mobile van.

CT LightSpeed Mobile Transit Preparation Procedure

This procedure must be completed to properly prepare your CT LightSpeed Mobile scanner for transport.

- 1. Remove accessories from the cradle and stow in cabinets for transport.**
- 2. Move the cradle out of the scanner to full out home position and raise the table to it's full up position.**
- 3. At the Operator's Console, select the pink "Shutdown" button in the desktop switching area. The system will ask you "Is this a shutdown for transit preparation?". Select YES.**
- 4. The system will prompt you to do the following:**
 - A. Install the Table Transport Support Stick at the foot end of the table.**
 - B. Tilt the gantry completely to full backward limit.**

Once you have completed these steps, the system will verify them. When recognized as completed, the system will complete shutdown of the software.
- 5. Verify that the red "Transport Warning Lamp" is not illuminated. This lamp is on the outside of the coach, near the front, on the driver's side. If the lamp is ON, the gantry is not completely tilted backwards.**
- 6. Once the gantry is tilted to full backward limit, the table is secured, and the system software has completed shutdown, you may power off the system at the Operator's Console.**
- 7. Make sure that the keyboard, mouse, trackball, and chair are all secured with their appropriate restraints.**
- 8. If your system has the Cardiac Scoring option with an EKG monitor, secure the monitor following the coach manufacturer's instructions.**
- 9. Follow the van manufacturer's instructions to prepare the van for transportation.**

CT LightSpeed Mobile Set-Up Procedure

This procedure must be completed to properly prepare your CT LightSpeed Mobile system for scanning.

- 1. Follow the van manufacturer's instructions to set up and level the van.**
- 2. Insure that the system power is turned on. The two green lights on the main power panel should be lit. If not, follow the "Start-Up" instructions posted near the power panel.**
- 3. Turn on the power switch on the front of the Operator's Console. The system will start to boot up.**
- 4. Remove the Table Transport Stick from under the front of the table. Stow it for use when transporting.**
- 5. After the system software has booted up, tilt the gantry to the "Zero Degree" position. Lower the table to a desired height for the first patient loading.**
- 6. Perform the "Daily Prep" procedures in the Operator's Manual. Begin patient scanning.**

A5 Addendum for all CT 9800 Mobile Systems

A5.1. General. The CT 9800 Mobile Program is no longer active and the CT 9800 scanners are scheduled for “End-of-life” (EOL). An EOL letter has been sent to all CT 9800 owners. Should the need arise for a CT 9800 Mobile installation, this request should be made to GEMS CT Engineering, Mobile Program Leader. GEMS CT Engineering will work with a suitable vendor for this installation on an as-needed, ad hoc basis to insure that a safe and serviceable installation can be accomplished.

A.6, Checklists

A.6.1 Used Vehicle, Phase-1 (Preliminary) Checklist

This checklist is to be accomplished by the GEMS Sales Rep. and forwarded to:

GE Medical Systems, CT Engineering
Attn: Mobile Product Team Leader
Mail Station W-1140
Box 414
Milwaukee WI 53201

Customer Name, Address, and Ph#: _____

Sales Rep Name: _____ Aspen #: _____

Model of CT to be sold: _____ Options: _____

Expected Delivery Date: _____

Please answer the following questions or place an "X" in the appropriate column:

	<u>Yes</u>	<u>No</u>	<u>N/A</u>
1. Does the mobile vehicle have a gross weight capacity of 40,000 lbs or more?	_____	_____	
2. Do the rear bogies or axles have an "Air-Bag" suspension system?	_____	_____	
3. If this is a trailer type unit, does the front king-pin have an air-bag?	_____	_____	_____
4. Will the customer be using the on-board generator to provide power for scanning (vs scanning only while on shore-power)?	_____	_____	
5. Does this van or coach have a back-up power generator on-board?	_____	_____	
6. Please provide the make, model, and age of vehicle: _____ _____			
7. Please state the dimensions of the vehicle: (Length, Width, and Height)			

8. Please state the dimensions of the scan room: (Length, Width, and Height)

9. Please state the dimensions, location, and description of any service entrances in the scan room.

10. Please provide a brief description of the vehicle or any other items you think are significant for this project: _____

A.6.2 Used Vehicle, Phase-2 Checklist.

Process: This checklist is to be accomplished by the Mobile Van Service Center (Vendor) prior to installation of the CT system and forwarded to:

GE Medical Systems, CT Engineering
Attn: Mobile Product Team Leader
Mail Station W-1140
Box 414
Milwaukee WI 53201

Please answer the following questions after reviewing this document, vehicle and scanner documentation, the CT Mobile system Maintenance Manual, and after inspecting the vehicle

1. Is there anything apparent that would deem that the vehicle is not road-worthy (conforms to all applicable DOT or local regulations)? See section 2-1. Note: A DOT inspection will be done prior to scanner installation.

Yes _____ No _____ Comments: _____
_____.

2. Is the air conditioning system and insulation adequate to maintain the environment? See section 2-2.

Yes _____ No _____ Comments: _____
_____.

3. Is the back-up power generator adequate and in good condition? See section 3.

Yes _____ No _____ Comments: _____
_____.

4. Is the electrical system adequate and in good condition? See section 3.

Yes _____ No _____ Comments: _____
_____.

5. Does the frame and chassis appear to be constructed in a way to properly support the necessary gantry and table mounting system?

Yes _____ No _____ Comments: _____
_____.

6. Is there adequate below-floor ducting for all cable runs between the gantry, the control room, and the PDU area?

Yes _____ No _____ Comments: _____
_____.

7. Does the suspension and king-pin appear in good condition ? (Should the project be approved, this will also need to be inspected and certified by a qualified service center as part of DOT inspection.)

Yes _____ No _____ Comments: _____
_____.

8. Does the scan room have adequate radiation shielding? (See specifications in this document, paragraph 6.4.).

Yes _____ No _____ Comments: _____
_____.

9. Does the vehicle have slide-outs to provide an enclosed environment for service?

Yes _____ No _____ Comments: _____
_____.

10. Please use your professional knowledge regarding mobile CT systems in identifying any other areas of concern (attach your checklist, inspection results, or criteria):

Comments:

A.6.3 Used Vehicle, Phase-3 Checklist.

The following checklist is to be used by the van service vendor as an installation checklist.

Step 1. Unpack and inventory scanner materials. Contact GEMS if anything is missing.

Step 2. Schedule subsystem Inspections:

- D.O.T. Inspection (Suspension, tires, wheels, road-worthiness/legal)
- Air Conditioning system
- Electrical systems
- Generator
- Radiation shielding
- Slide-outs

Perform upgrades as needed.

Step 3. Prep unit for CT equipment installation per GE specifications.

- Cut and mount steel base-plates for table and gantry.
- Fabricate brackets for OC base, CRT, & PDU isolators/dampening system.
- Determine equipment locations per drawings.
- Produce cable ducts and access holes for cable runs. Isolate or remove any sharp edges.

Step 4. Load and mount system components into van per GE specifications.

- Drill and tape gantry mounts and install bolts and spacers.
- Mount PDU with isolators.
- Anchor O.C. to floor and wall.
- Mount monitor(s).
- Position, table and gantry.

Step 5. Route and terminate all system cables. Install the Mobile Kit provided by GE (If applicable).

Step 6. Level the van first, then level the gantry and table. Perform a table/gantry alignment and secure them to the floor.

Step 7. Install and test a two-bulb transit warning system and connect to the GE CT System.

Step 8. If this is a unit which will not be scanning on back-up power, connect one switch contact of the input power switch into the Room Door Interlock circuit of the CT System. (Refer to section 3.5 of this Document). Connect door interlock circuit to the CT System (See Installation Manual).

Step 9. Install the Power Switch/Breaker Box within 1.5 meters of the CT PDU per GE specifications. Connect power cables to CT system.

Step 10. Install table Transit Bracket ("Ski" bracket), D-Rings into the floor, and cinch-straps. Place a laminated copy of the Transit Preparation instructions on the scan-room wall. Make sure that the bracket has the appropriate warning label. (Refer to Sections 4.6.5 and A2.6 of the Requirements Document.)

Step 11. Other items which must be accomplished in no particular order:

- Produce and attach warning label for transport warning lamp.
- Install telephone line and RJ-45 jack near or behind O.C.
- Install camera or other optional devices.
- Label gantry service tools and secure in storage compartment or back of scan room.
- Line phantom cabinet with foam rubber and secure phantoms.

Step 12. Complete the GEMS CT Design Compliance Checklist. Forward a copy to The GEMS Global Mobile Manager.

Step 13. Schedule the unit for transport to the GEMS Mobile facility for inspection and calibration. In Europe, contact GEMS to arrange for final inspections and calibration.



GE Medical Systems
